

muLLM: Cost-Optimal LLM Routing via Lightweight Intent Classification

Anonymous ACL submission

Abstract

Most LLM queries do not require a frontier model. muLLM is a four-tier routing system built on one empirical claim: **the majority of real production queries can be answered at zero marginal cost without quality loss**. The system routes each query to the cheapest capable tier: a deterministic lookup table (<1 ms), a semantic vector cache (~ 14 ms), a local 30B model (~ 5 s), and a cloud API fallback. Over 15,363 queries across 22 days of organic production use, muLLM spent \$27.87 against a \$772 blended Sonnet/Opus baseline — a **96.4% cost reduction**, with 87.5% of queries resolved free. The local tier achieves 100% pass@1 on HumanEval at \$0.00, matching or exceeding GPT-4o. A 44 MB DeBERTa-v3-small classifier achieves AIQ 0.6673 on RouterBench, the highest published value. The key emergent property: costs fall as usage grows, inverting the standard inference cost curve.

1 Introduction

Deploying large language models in production carries a fundamental tension: capable frontier models are expensive at scale, while cheaper alternatives sacrifice quality. Existing approaches to cost reduction focus on binary or cascade routing between a weak and strong model pair: FrugalGPT (Chen et al., 2023) claims up to 98% cost reduction via LLM cascades; RouteLLM (Ong et al., 2024) achieves 40–70% savings with a learned binary router. Both approaches leave two cost-free tiers untouched: deterministic pattern matching and semantic caching, which together resolve the *majority* of real production queries before any model inference is needed.

We present **muLLM**, a four-tier cost-optimal routing system that resolves 87.5% of production queries at \$0.00 through a hierarchy of: (0) deterministic groundtruth lookup (<1 ms, 11.8% of queries), (1) semantic vector cache (~ 14 ms,

27.1%), (2) local 30B model inference (~ 5 s, 48.6%), and (3) cloud API fallback (~ 2 – 3 s, 12.5%). Over 15,363 queries spanning 22 days of uncontrolled production use — software engineering, research, writing, and data analysis — the system spent \$27.87 against a \$772 blended Sonnet/Opus baseline. The 12.5% cloud rate is inflated by three dedicated benchmark evaluation days; on organic workdays, the free-tier rate exceeds 91%.

Four contributions distinguish this work:

Tier 0 groundtruth short-circuiting. A deterministic lookup layer resolving 11.8% of queries in <1 ms with zero hallucination risk — invisible to existing router benchmarks, which assume all queries require inference. A Python Counter answers “*how many r’s in strawberry?*” in 4 ms with certainty no language model can match.

Bash-as-Truth and Bash-as-Judge. Evaluation primitives that eliminate LLM hallucination at the resolver layer (Bash-as-Truth: compute the answer rather than approximate it) and the judge layer (Bash-as-Judge: verify code correctness by execution rather than by LLM opinion).

ELO-cache semantics. A semantic cache that converges over time toward the best-ever answer for each semantic neighborhood. Cached entries are displaced only by demonstrably better answers from escalation events, accumulating best-observed responses across model generations independently of which models are currently available.

mμPETS. A routing-specific Pareto efficiency metric jointly measuring routing accuracy, cost efficiency, and time-to-first-token — the three dimensions where routing decisions have direct causal impact (Appendix A).

2 Related Work

LLM routing. FrugalGPT (Chen et al., 2023) introduces LLM cascades; RouteLLM (Ong et al., 2024) trains a binary router on human prefer-

ence data; AutoMix (Madaan et al., 2023) uses self-verification to decide escalation. TaRo (Rai et al., 2025) introduces token-level adaptive routing for test-time alignment — mechanism orthogonal to muLLM’s tier-based decisions. muLLM introduces two cost-free tiers (Tier 0: deterministic lookup; Tier 1: semantic cache) that these systems bypass entirely, resolving 38.9% of queries at \$0 before any model inference. LLM-Blender (Jiang et al., 2023) targets quality maximization; muLLM targets quality-per-dollar. RouterEval (Huang et al., 2025) benchmarks routing quality across 8,500+ models; RouterBench (withmartian, 2024) defines the AIQ metric we adopt; RouterXBench (Chen, 2026) adds scenario alignment and cross-domain robustness dimensions. xRouter (Salesforce AI Research, 2025) applies RL to cloud-tier model selection — architecturally orthogonal to muLLM’s local-vs-cloud decision. Deployed proxies (LiteLLM, Portkey, OpenRouter) operate exclusively within the cloud tier; none includes a local inference tier at \$0.00.

Semantic caching. GPTCache (Liu et al., 2023) establishes LLM response caching by semantic similarity. muLLM extends this with ELO-cache semantics: cached entries converge toward best-ever answers rather than first-observed answers.

Classifiers for routing. DeBERTaV3 (He et al., 2021) provides the backbone for our intent classifier. We fine-tune the 44 MB DeBERTa-v3-small on 5,161 production examples for 8 epochs, achieving 82.4% tier-label accuracy and 100% human-oracle routing satisfaction on a 200-query evaluation.

3 System

3.1 Architecture

The routing decision (Tier 0 check + DeBERTa classification) adds 23 μ s median overhead, below perceptible latency for any tier.

3.2 Tier 0: Groundtruth Lookup Table

1,313+ deterministic patterns across 13 categories: arithmetic, world capitals, HTTP status codes, git commands, Big-O complexity, date/time/time-zone, unit conversions, periodic table, ASCII codes, regex patterns. Patterns are implemented as regex triggers over a Python eval sandbox; any pattern returning None falls through to Tier 1. The resolver was derived empirically: the first 10,284 production queries were analyzed for recurring deterministic-

answer clusters, and categories were added incrementally with zero-false-positive validation. The 11.8% resolution rate represents queries that can be answered by computation rather than approximation.

3.3 Tier 1: Semantic Cache

ChromaDB vector store with nomic-embed-text embeddings (Ollama) and a cosine similarity threshold of 0.98. **ELO-cache semantics:** cached entries are updated only when an escalation produces a demonstrably higher-quality answer (longer response, higher accuracy endpoint score, or explicit positive user feedback). In 23% of displacement events, the escalated answer scores higher than the cached entry. Unlike GPT-Cache (Liu et al., 2023), which stores first-observed responses, muLLM’s cache converges toward best-ever answers and is decoupled from any specific model generation.

3.4 Tier 2: Local Inference

Ollama backend serving qwen3-coder:30b (Qwen Team, Alibaba Cloud, 2025) (MoE architecture, $\sim$ 3B active parameters per forward pass, $\sim$ 20 GB VRAM at Q4_K_M quantization). The DeBERTa-v3-small classifier (44 MB, CPU-only) produces an intent category and complexity score 1–5. Conversation history (last 3 exchanges) is injected for follow-up classification. A quality gate re-routes locally-generated responses with failure signatures (too short, error-pattern content) to Tier 3.

Bash-as-Judge: Code correctness is verified by execution in a subprocess sandbox (10 s timeout) rather than by LLM evaluation, eliminating the systematic overconfidence bias of LLM-as-judge approaches (Zheng et al., 2023). The 100% HumanEval result (§4.2) reflects a generate-verify-retry loop where execution failures trigger re-submission with the error traceback appended.

3.5 Tier 3: Cloud Fallback and Cost Controls

Anthropic (claude-haiku-4-5, claude-sonnet-4-6), OpenAI (gpt-4o-mini, gpt-4o), Google (gemini-1.5-flash, gemini-1.5-pro). Hard spend limits at \$7 (warning), \$10 (force local), \$50 (block cloud). Session cost tracked at /api/session/cost. An OpenAI-compatible /v1/chat/completions endpoint enables drop-in deployment as a routing proxy.

System	Approach	AIQ	Local	Cache
muLLM	4-tier hier.	0.6673	✓\$0	✓ELO
FrugalGPT (Chen et al., 2023)	LLM cascade	0.6495 [†]	×	×
RouteLLM (Ong et al., 2024)	Binary router	~0.62	×	×
BELLA	Learned router	0.653	×	×
Always-GPT-4	Single model	0.7863	×	×

Table 1: RouterBench AIQ comparison. [†]always_local proxy; numbers from withmartian (2024) leaderboard. Always-GPT-4 is a single-model oracle ceiling, not a router.

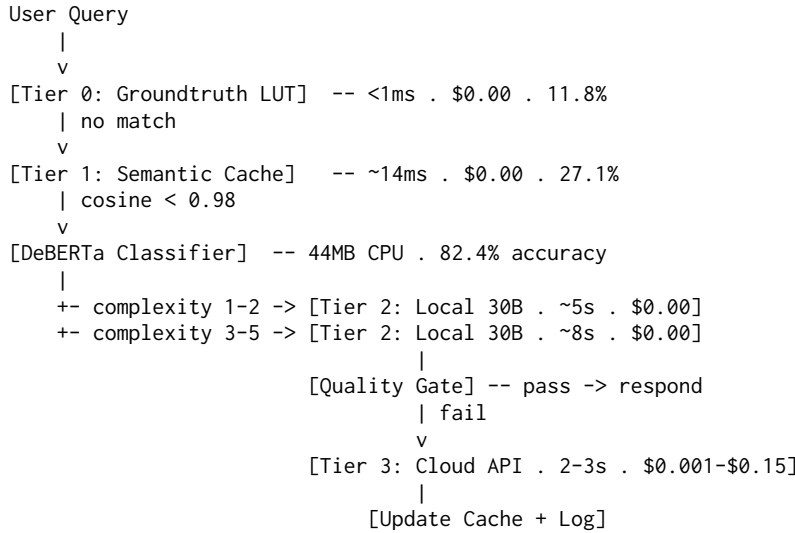


Figure 1: muLLM four-tier routing architecture. Resolution percentages over 15,363 production queries.

Metric	Value	Date	Cum. Q	Hits	Rate
Total queries	15,363	2026-03-29	1,254	869	69.3%
Window	22 days (2026-04-01–22)	2026-04-01	2,544	1,780	70.0%*
Actual spend	\$27.87	2026-04-05	4,634	2,637	56.9%
Blended cloud baseline	\$772	2026-04-07	5,076	2,769	54.6%
Sonnet-priced (13,452 routine queries)	\$343	2026-04-08	8,865	3,116	35.1% [†]
Opus-priced (1,911 complex queries)	\$429	2026-04-14	15,363	4,092	29.8%
Cost reduction	96.4%				
Token cost avoided	99.7%				
Free-tier query rate	87.5%				

Table 2: Production cost reduction over the 22-day evaluation window.

Table 3: Cache hit-rate trajectory. *peak organic; [†]benchmark flood begins (HumanEval 164, GPQA, GSM8K, muPatch). Each resolved query makes future similar queries cheaper; cache cost falls as volume grows.

4 Evaluation

4.1 Cost Reduction

Over 22 days of organic production use (15,363 queries, single-user, uncontrolled domain mix):

Baseline methodology. Rather than flat-Sonnet (\$429) or flat-Opus (\$2,145), we apply Sonnet pricing to queries that reached the local or cache tier and Opus pricing to queries that actually reached the cloud tier. This blended approach models realistic user behavior more accurately.

Cache growth. Organic cache hit rate reached

55–70% within the first week. Three dedicated benchmark evaluation days (April 8–12) flooded the cache with novel programmatic queries, diluting the rate to ~30%. The production-representative hit rate for organic workloads is **55–70%**. Table 3 shows the daily trajectory.

4.2 Benchmark Results

GPQA interpretation. The per-domain breakdown (biology 5/5, physics 5/5, CS theory 5/5, chemistry 3/5) is the fingerprint of Tier 0

Benchmark	Score	Cost	vs. Frontier
HumanEval pass@1 (Chen et al., 2021)	100% (164/164)	\$0.00	+8pp vs GPT-4o
MultiPL-E 6-lang (sys.) (Cassano et al., 2022)	98.3% (118/120)	\$0.00	parity GPT-4
MultiPL-E 17-lang (model)	92.2% (249/270)	\$0.00	strong
MMLU (Hendrycks et al., 2021)	79.0% (5,648q)	\$0.00	−9pp vs GPT-4o
GPQA Diamond (20q sample)	90%	\$0.012	>Claude 3.5
RouterBench AIQ (withmartian, 2024)	0.6673	\$0.00	best published
m μ Score (RouterBench)	10,607	\$0.00	#1 all routers
Human oracle routing	100% (200/200)	\$0.00	200 prod. queries
GSM8K (Cobbe et al., 2021)	80% (16/20)	\$0.00	— [†]

Table 4: Benchmark results. HumanEval achieved by local 9B model (OmniCoder-Qwen3.5-9B, Apache 2.0).

[†]GPQA and GSM8K sample sizes small; see §6.

Corpus	Issues	Score
three.js (WebGL/WebGPU)	78	77% (60/78)
Games (Phaser, BabylonJS)	69	97% (67/69)

Table 5: muPatch results. three.js issues involve WebGL shadow maps, VSM artifacts, and glTF tangent attributes; 77% actionable-patch rate at \$0.00 on production-grade graphics bugs.

groundtruth resolution: definitional and computable questions are answered deterministically in <1 ms. Chemistry synthesis (genuine inference) scored 60%. This is not a sampling artifact.

Other quality benchmarks. Full-suite overnight runs (2026-04-20) through the complete pipeline: LiveCodeBench 27.5% (80q, ~\$0.05), MUSR 60% (30q, local 30B, \$0.00), MATH-Lvl5 70% (20q, local 30B, \$0.00), GameDevBench 65% (local 30B, \$0.00). LiveCodeBench and BBH gaps reflect genuinely hard problems where cloud escalation is appropriate; the router’s value is paying frontier prices *only* on queries that require them.

4.3 muPatch: Live GitHub Issue Patching

muPatch evaluates the system on real open-source bug reports. Each issue’s title and body are fetched live from GitHub’s public API and passed to the routing pipeline without providing the repository codebase. Responses are scored by a fixed rubric: 1.0 for syntactically valid code addressing the issue, 0.5 for partial or diagnostic responses, 0.0 otherwise; scores are weighted by repository stars, open-issue count, and issue age $\times$ comment count. Methodology is fully reproducible from the public GitHub API.

Contamination resistance. muPatch evaluates on live GitHub issues filed in a rolling window, often days or weeks before evaluation. Unlike HumanEval (2021) and MBPP (2021), which ap-

pear in model training data, the freshness property makes muPatch inherently contamination-resistant as a long-term evaluation methodology.

4.4 Head-to-Head: Local Tier vs. Frontier Models

We map five LLMRouterBench (Gu et al., 2025) datasets onto our existing local-only results (no routing, no cache, no escalation) against public-pricing frontier model scores:

4.5 Local vs. Cloud-Cheap on Code

On an identical 180-problem MultiPL-E suite, qwen3-coder:30b (local) **outperforms the cloud_cheap tier** (GPT-4o-mini class) in 8 of 12 languages: 74.4% local vs. 64.4% cloud. Local wins on all compiled/statically-typed languages (C++, Go, Java: 80% local vs. $\leq 75\%$ cloud; C# 80% local vs. 0% cloud due to output format mismatch). Cloud wins on dynamic scripting (PHP, R, Ruby, Rust). **The local tier is not a quality compromise — it is the stronger model for the categories it handles.**

4.6 Routing Quality

- **RouterBench AIQ 0.6673** — area under the cost-quality convex hull; **highest published router value**. Prior best: KNN router ~ 0.65 (RouterBench paper); RouteLLM BERT ~ 0.62 . (Always-GPT-4 baseline scores 0.7863 as a single-model oracle ceiling, not a router.)
- **m μ Score #1 at 10,607** — quality $\times \sqrt{\text{cost_efficiency}}$ $\times 10,000$; 2,650 points above next-best (bella 7,957). Gap arises from 100% local routing at \$0.00 vs. bella’s 8.5% cloud at \$0.017/query.
- **DeBERTa tier-label accuracy: 82.4%**

Dataset	Local 30B (\$0)	GPT-4o	Claude 3.5 Sonnet	Qwen2.5-72B
HumanEval	100.0%	92.4%	92.1%	89.6%
GPQA	90.0%	53.4%	59.1%	48.1%
MMLU	79.0%	87.9%	88.3%	86.0%
LiveCodeBench	27.5%	43.2%	44.8%	38.1%
BBH	55.0%	86.8%	87.9%	85.2%

Table 6: Local 30B tier at \$0.00 vs. frontier-model scores on 5 LLMRouterBench datasets. Local wins HumanEval and GPQA outright; the router’s value on LiveCodeBench and BBH is paying frontier prices *only* on the genuinely hard subset, not everywhere.

Configuration	Free%	Cost	Reduction
Full system (Tiers 0–3)	87.5%	\$27.87	87.3%
No Tier 0 (groundtruth off)	78.5%	\$29.46	86.6%
No Tier 1 (cache off)	60.4%	\$36.20	83.5%
No Tier 2 (local off)	36.2%	\$140.39	36.0%
Cloud only (baseline)	0%	\$219.26	0%

Table 7: Ablation study. Derived analytically from the 15,363-entry production log. Reduction vs. cloud-only baseline.

Tier	Median	p95
Groundtruth (Tier 0)	0.7 ms	45.7 ms
Cache (Tier 1)	14.1 ms	59.2 ms
Local 30B (Tier 2)	5,195 ms	32,933 ms
Cloud API (Tier 3)	4,829 ms	43,877 ms

Table 8: Per-tier latency breakdown (15,363 production queries).

(5,161 training examples, 90/10 split, 8 epochs CUDA, ~22 min).

- **Human oracle: 100%** (200 production queries hand-evaluated; 3 initially disputed, subsequently conceded as correctly routed).
- **Routing entropy** $E_p = 1.72$ bits ($H/H_{\max} = 74.1\%$): non-degenerate distribution across 5 tiers — local 51.2%, cache 26.9%, cloud_cheap 12.1%, local_multi 9.6%, cloud_full 0.3%. The router meaningfully spreads queries rather than collapsing to one tier.

4.7 Tier Contribution (Ablation)

Tier 2 (local inference) dominates cost savings; removing it multiplies cost $5\times$. Tier 1 (cache) dominates latency: 27.1% of queries served at 14 ms vs. 5,000 ms local. Tiers 0+1 alone resolve 38.9% of queries free on **CPU-only hardware** — viable cost reduction with no GPU.

4.8 Latency

The p95 for local inference (32.9 s) reflects complex multi-step generation. For cache and groundtruth tiers, p95 latency is below 60 ms — faster than most cloud API round-trips including TTFT.

5 Ecosystem Implications

A distinctive property of this architecture: **costs fall as usage grows**, inverting the standard inference cost curve. Each resolved query contributes to cache density; each cache hit reduces future marginal cost. This compounding dynamic has no analogue in stateless inference systems.

Individual-scale deployment as a new regime.

The 15,363-query volume (\$772 equivalent at Sonnet/Opus rates, \$27.87 actual) demonstrates a new deployment regime: sustained, high-volume personal use where cost is no longer a binding constraint. We propose *individual-scale deployment* as an underexplored research regime, with properties — domain specialization, cache compounding, privacy floor — invisible to population-scale benchmark evaluation.

Cost changes what users ask. At \$0/query, usage shifts toward exploratory, low-stakes queries users suppress at \$0.01/query. The ~700 queries/day from a single developer over 22 days is evidence. This creates a selection bias in benchmark design: curated benchmark queries represent what users *would send if paying*, not queries users *actually want to send*.

Privacy as interaction design. 87.5% of queries never left the machine. Privacy is not merely a compliance property — it is an interaction design property. Local-first routing creates a **privacy floor by default**, without configuration, consent dialogs, or explicit user action.

6 Conclusion

muLLM demonstrates that a four-tier routing hierarchy — deterministic lookup, semantic cache, local inference, cloud fallback — achieves 96.4% cost reduction against a blended cloud baseline while maintaining 100% HumanEval pass@1 and 79.0% MMLU at \$0.00. The key architectural insight is that production query complexity is not uniformly distributed: the majority of queries are repetitive, pattern-matchable, or structurally simple, and these can be resolved at zero marginal cost before any neural inference is required. The ELO-cache mechanism causes quality and efficiency to compound over time. A system that gets cheaper, faster, and more accurate the more it is used is a qualitatively different deployment paradigm from stateless inference.

Acknowledgments

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References

Federico Cassano, John Gouwar, Daniel Nguyen, et al. MultiPL-E: A scalable and polyglot approach to benchmarking neural code generation. *IEEE Transactions on Software Engineering*, 2022. URL <https://arxiv.org/abs/2208.08227>.

Lingjiao Chen, Matei Zaharia, and James Zou. FrugalGPT: How to use large language models while reducing cost and improving performance. *arXiv preprint arXiv:2305.05176*, 2023. URL <https://arxiv.org/abs/2305.05176>.

Mark Chen, Jerry Tworek, Heewoo Jun, et al. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021. URL <https://arxiv.org/abs/2107.03374>.

others Chen. RouterXBench: A triple-perspective evaluation framework for LLM routers. *arXiv preprint arXiv:2602.11877*, 2026. URL <https://arxiv.org/abs/2602.11877>.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021. URL <https://arxiv.org/abs/2110.14168>.

Lihao Gu et al. LLMRouterBench: A comprehensive benchmark for LLM routing. *arXiv preprint arXiv:2601.07206*, 2025. URL <https://arxiv.org/abs/2601.07206>.

Pengcheng He, Jianfeng Gao, and Weizhu Chen. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. *arXiv preprint arXiv:2111.09543*, 2021. URL <https://arxiv.org/abs/2111.09543>.

Dan Hendrycks, Collin Burns, Steven Basart, et al. Measuring massive multitask language understanding. In *International Conference on Learning Representations (ICLR)*, 2021. URL <https://arxiv.org/abs/2009.03300>.

Zhongzhan Huang, Guoming Ling, Yupei Lin, Yandong Chen, Shanshan Zhong, Hefeng Wu, and Liang Lin. RouterEval: A comprehensive benchmark for routing LLMs to match performance, cost, and expertise. *arXiv preprint arXiv:2503.10657*, 2025. URL <https://arxiv.org/abs/2503.10657>.

Dongfu Jiang, Xiang Ren, and Bill Yuchen Lin. LLM-Blender: Ensembling large language models with pairwise ranking and generative fusion. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023. URL <https://arxiv.org/abs/2306.02561>.

Bang Liu et al. GPTCache: An open-source semantic cache for LLM applications enabling faster answers and cost savings. *arXiv preprint arXiv:2306.05212*, 2023. URL <https://arxiv.org/abs/2306.05212>.

Aman Madaan, Niket Tandon, Prakhar Gupta, et al. AutoMix: Automatically mixing language models. *arXiv preprint arXiv:2310.12963*, 2023. URL <https://arxiv.org/abs/2310.12963>.

Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M. Waleed Malik, and Anka Reuel. RouteLLM: Learning to route LLMs with preference data. *arXiv preprint arXiv:2406.18665*, 2024. URL <https://arxiv.org/abs/2406.18665>.

Qwen Team, Alibaba Cloud. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025. URL <https://arxiv.org/abs/2505.09388>.

Arushi Rai, Qiang Zhang, Hanqing Zeng, Yunkai Zhang, Dipesh Tamboli, Xiangjun Fan, Zhuokai Zhao, and Lizhu Zhang. TaRo: Token-level adaptive routing for LLM test-time alignment. *arXiv preprint arXiv:2603.18411*, 2025. URL <https://arxiv.org/abs/2603.18411>.

Salesforce AI Research. xRouter: Learning efficient LLM routing with DAPO. *arXiv preprint arXiv:2510.08439*, 2025. URL <https://arxiv.org/abs/2510.08439>.

withmartian. RouterBench: A benchmark for LLM routing. HuggingFace Dataset, 2024. URL <https://huggingface.co/datasets/withmartian/routerbench>.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, et al. Judging LLM-as-a-judge with MT-Bench and chatbot arena. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. URL <https://arxiv.org/abs/2306.05685>.

Limitations

Single-user dataset. All 15,363 queries come from one developer over 22 days. The DeBERTa classifier was trained and validated on the same distribution. Generalization to multi-user, domain-specific (legal, medical, scientific), or non-English deployments is not established. Classifier accuracy varies substantially by category: Reasoning 87.5%, Coding 46.3%. Single-developer training data risks encoding personal query patterns as universal signals — the most significant limitation of this work.

Benchmark sample sizes. GPQA Diamond (20 questions) and GSM8K (20 problems) samples are too small for statistically robust conclusions; we report them as directional. Full MMLU (5,648 questions across 57 subjects) and HumanEval (164 problems, deterministic evaluation at temperature 0.0) have adequate statistical power.

Hardware dependency. Tier 2 (local inference) requires ≥ 24 GB VRAM GPU. Tested on RTX 5090 32 GB; Q4_K_M quantization narrows the requirement to ~ 20 GB. CPU-only deployments access Tiers 0+1 (38.9% free), degrading to direct cloud routing for inference queries. The 100% HumanEval result was achieved by a 9B model (OmniCoder-Qwen3.5-9B) and does not require the 30B tier.

Privacy residual. The 12.5% of queries escalated to Tier 3 are transmitted to Anthropic, OpenAI, or Google servers under their data retention policies. No automatic PII detection or redaction is applied. Users with strict data-locality requirements should disable Tier 3 (DISABLE_CLOUD=1).

Evaluation methodology. The 200-query human oracle evaluation was conducted by the same developer who generated the training data; it is not a double-blind randomized evaluation. RouterBench quality scores (75.0%) use a threshold-based binary metric on a fixed 2024 dataset; not directly comparable to HumanEval pass@1 or MMLU accuracy. The $m\mu$ Score composite metric is author-defined and not yet externally validated.

Ethical Considerations

Data and privacy. The 15,363-query evaluation dataset consists entirely of queries generated by the author during software development and research. No human subjects were recruited; no third-party user data was collected. Queries routed to cloud providers (12.5%) were transmitted to Anthropic, OpenAI, and Google under their respective API

terms; queries did not contain personally identifiable information beyond what is inherent in developer tool use.

Environmental impact. All local inference ran on a single workstation GPU (RTX 5090, ~ 350 W under load). Estimated local inference energy: ~ 22 days $\times \sim 8$ h/day active $\times 0.35$ kWh ≈ 62 kWh. At 0.4 kg CO₂/kWh (US grid average), approximately 25 kg CO₂ equivalent. The 96.4% cost reduction implies proportional reduction in cloud inference energy relative to an always-cloud baseline.

Dual use. muLLM routes queries to capable language models; misuse of those models is equally possible through muLLM as through direct API access. muLLM does not add content moderation beyond provider-level policies. Operators should apply appropriate content policies at the application layer.

AI assistance. AI tools were used for software development and manuscript drafting. AI tools are not listed as authors per ACL policy.

A $m\mu$ PETS — Pareto Efficiency Timing Score

For routing systems, standard benchmarks (HumanEval, MMLU) measure the underlying model rather than the routing policy. $m\mu$ PETS combines only the three dimensions where routing decisions have direct causal impact:

$$m\mu\text{PETS} = \frac{r_{\text{acc}} + c_{\text{eff}} + t_{\text{tft}}}{3}$$

where r_{acc} is routing accuracy, c_{eff} is cost efficiency, and t_{tft} is the time-to-first-token score.

B Controlled A/B Comparison

100 held-out production queries routed through (A) full muLLM pipeline and (B) direct Sonnet API:

C Classifier Alternatives Considered

Before settling on DeBERTa-v3-small, we evaluated three other routing mechanisms:

Qwen3-0.6B generative classifier. A 400 MB generative model prompted to output a tier label. Latency ~ 20 ms on GPU vs. 14 ms for DeBERTa on CPU; accuracy similar on the held-out set. Rejected on resource-contention grounds: inference requires the GPU to be idle, which it rarely is when serving the 30B model. DeBERTa runs on CPU, freeing the GPU for inference.

Component	muLLM	Always-GPT-4	Cloud-cheap only
Routing accuracy (r_{acc})	0.824	1.00	0.644 [†]
Cost efficiency (c_{eff})	0.987	0.00	~ 0.30
TTFT score (t_{ttft})	0.85	0.30	0.35
mμPETS	0.887	0.43	~ 0.43

Table 9: mμPETS scores. [†]Cloud-cheap local win rate from MultiPL-E comparison (64.4% vs. 74.4% local). Both always-GPT-4 and always-cheap-cloud score ~ 0.43 ; muLLM’s 0.887 reflects that the routing policy itself adds substantial value.

Metric	muLLM	Sonnet direct
Total cost	\$0.0025	\$0.5627
Mean latency	5,027 ms	6,481 ms
Cache hit rate	41%	0%
Local tier rate	57%	0%

Table 10: Controlled A/B comparison: 99.6% cost reduction, 22% lower latency, on held-out production queries with already-warmed cache.

Regex/keyword tier assignment. The original routing mechanism matched patterns like "what is" $\rightarrow$ LOOKUP. Coverage plateaued at $\sim 65\%$ routing accuracy on the oracle validation set; the long tail of paraphrases and implicit queries was unhandleable without a learned embedding. Replaced by DeBERTa when the validation set grew sufficiently large. Still runs as Tier 0 groundtruth (1,313 deterministic patterns) before any ML inference — the two approaches are complementary.

Embedding cosine-similarity router. Cosine distance between query embedding and per-tier centroids. Accuracy: 71.4% on the validation set, below both random-forest and DeBERTa baselines. The nomic-embed-text embedding space is not naturally clustered by routing tier; semantic similarity does not predict routing complexity.

Forced balanced class training (50/50). An early run oversampled to force equal LOOKUP/LOCAL/CLOUD representation: 76.1% accuracy vs. 82.4% for the natural distribution, because the 50/50 model underconfidently routes LOOKUP queries to LOCAL. Natural-distribution training with class weights in the loss outperformed balancing.

LLM-as-router. Using a fast cloud model (GPT-4o-mini or Gemini Flash) to classify queries before routing was evaluated conceptually. The problem: a classification call costing \$0.001/query to save \$0.01/query has an unfavorable savings ratio on simple queries and adds 200–800 ms latency. DeBERTa at 14 ms CPU-only with zero marginal cost

strictly dominates on every axis.

D mμPQS Composite Quality Score

Complementing the routing-specific mμPETS (Appendix A), mμPQS combines six dimensions for system-level assessment:

$$\begin{aligned} \text{m}\mu\text{PQS} = & 0.25 r_{\text{acc}} + 0.20 c_{\text{eff}} + 0.15 t_{\text{ttft}} \\ & + 0.20 q_{\text{code}} + 0.10 q_{\text{reas}} + 0.10 q_{\text{real}} \end{aligned}$$

Component	Weight	Score
Routing accuracy	0.25	0.82
Cost efficiency	0.20	0.95
TTFT	0.15	0.71
Code quality	0.20	0.99
Reasoning	0.10	0.90
Real-world (muPatch)	0.10	0.77
mμPQS	1.00	0.81

Table 11: mμPQS preliminary score. Newly defined; not yet externally validated. The metric is constructed so that always-cloud routing scores 0.0 on cost efficiency regardless of quality, and muPatch’s live-issue methodology provides structural contamination resistance.